

AUTOMATED QUALITY INSPECTION ROBOT FOR WELDED WIRE MESH PRODUCTION

PROBLEM /NEED

Welded wire mesh requires inspection of mesh spacing, weld quality and coating uniformity. Manual inspection can be slow and inconsistent, and small defects may be missed. Inspection results also need to be recorded for traceability. The proposed system provides automated inspection while supporting integration with the production flow.

DESIGN REQUIREMENTS

- Controlled/indexed mesh transport
- Stable camera field of view
- Controlled LED illumination
- Automatic OK / NOT-OK decision
- Defect classification
- Inspection report generation
- Low-cost and modular prototype
- Production-line integration potential

PROPOSED SOLUTION

The proposed system uses an indexed-feed mechanism to move welded wire mesh to a predefined inspection position. A fixed camera captures the mesh under controlled LED illumination.

The image is transferred to a laptop running OpenCV, where perspective correction, thresholding, mesh-pitch measurement and weld-feature analysis are performed. After labelled inspection images are collected, an ML classifier can be used to classify the inspection result.

The proposed output categories are:

- **OK**
- **Spacing Defect**
- **Weld Defect**
- **Coating Defect**

The inspection result is stored with the image, defect information and timestamp to create a digital inspection record.

KEY IDEA

Indexed mesh feeding + controlled image acquisition + computer vision + defect classification = consistent and traceable mesh inspection.

SYSTEM CONCEPT

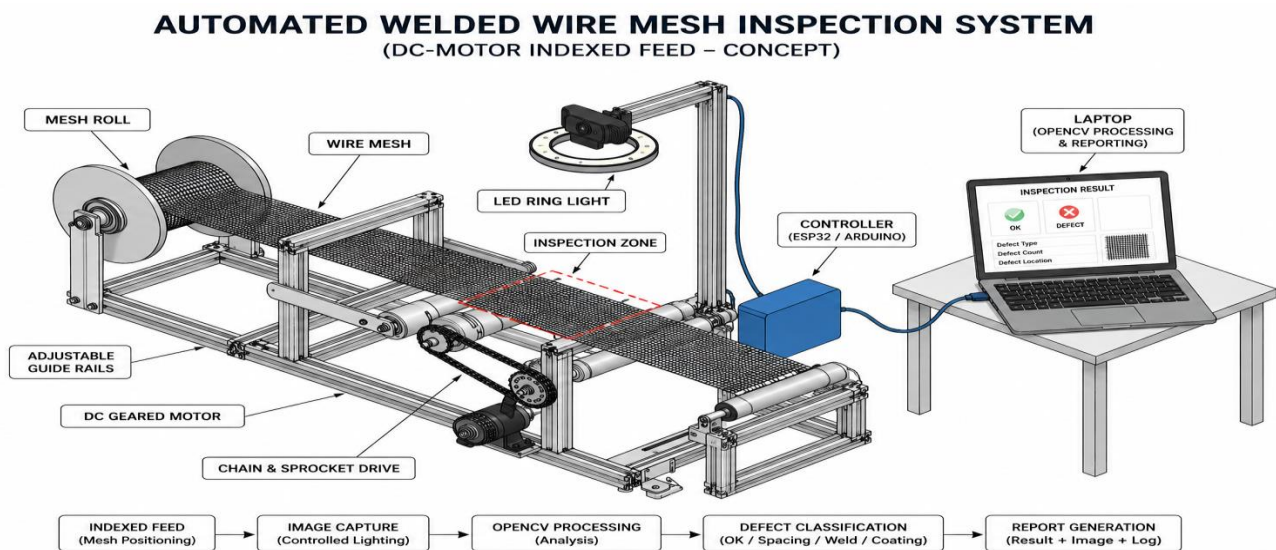


Figure 1. Conceptual visualization of the proposed automated welded wire mesh inspection system.

MECHANICAL SYSTEM

The proposed prototype uses a 40 × 40 mm aluminium profile frame to support the mesh transport and inspection assembly.

Proposed CAD/prototype dimensions:

- Base frame: 1800 × 900 × 500 mm
- Mesh working width: 450 mm
- Conveyor deck height: 500 mm
- Rollers: Ø60 × 500 mm
- Two idler rollers for guidance
- Adjustable guide rails
- Adjustable camera tower
- Camera-to-mesh distance: 650-700 mm

DRIVE SYSTEM

A 12 V DC geared motor drives the main roller through a chain and sprocket arrangement. Idler rollers and adjustable guide rails support and guide the mesh. The indexed-feed mechanism positions the mesh at the inspection location before image capture.

CAMERA & ILLUMINATION

A fixed camera is mounted above the inspection zone using an adjustable camera tower. Controlled LED illumination provides stable lighting during image acquisition.

ELECTRONIC SYSTEM

12 V DC Supply → Fuse + E-Stop → Motor Driver → 12 V DC Geared Motor

The ESP32 / Arduino provides motor control and can receive encoder or limit inputs where applicable. A USB camera connects to the laptop running OpenCV.

MAIN COMPONENTS

- Mechanical Components
- 40 × 40 mm aluminium profile frame
- Ø60 × 500 mm rollers
- 12 V DC geared motor
- Chain and sprocket drive
- Adjustable guide rails
- Adjustable camera tower

Vision & Illumination

- USB camera
- LED illumination
- Laptop with OpenCV

Control & Safety

- ESP32
- Motor driver
- 12 V DC power supply
- Fuse and emergency-stop (E-Stop)

CAD FIGURES

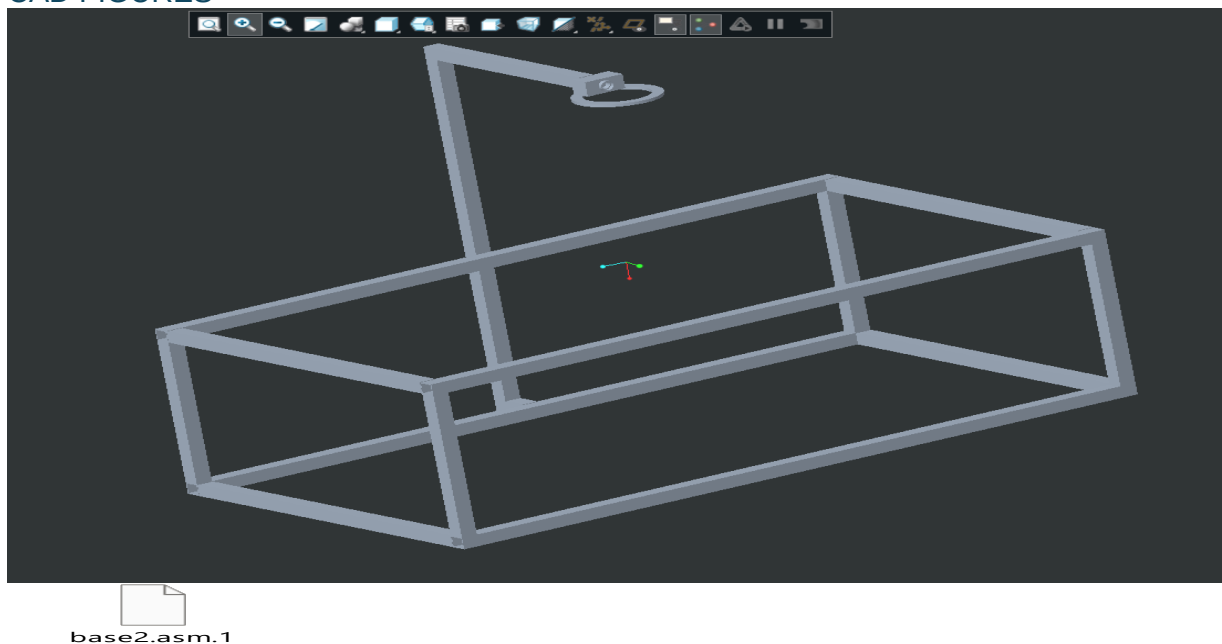


Figure 2. CAD model of the proposed automated welded wire mesh inspection system.

SYSTEM ARCHITECTURE

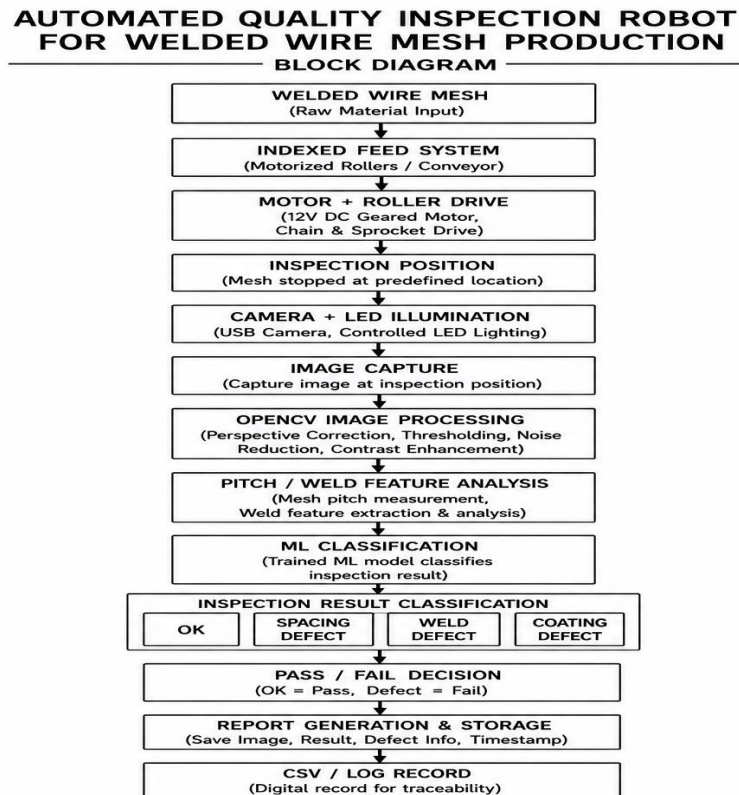


Figure 3. Functional block diagram of the proposed inspection and reporting workflow.

INSPECTION OUTPUT

- PASS / FAIL status
- Defect category
- Defect count
- Defect location
- Saved inspection image
- Timestamp
- CSV / log record

PROPOSED VALIDATION PLAN

Testing will use OK samples, controlled spacing-defect samples, controlled weld-defect samples and coating-defect samples, with multiple samples for each class.

Evaluation will consider:

Accuracy | Precision | Recall | False-reject rate | Repeatability under fixed lighting

EXPECTED BENEFITS

1. Reduced manual inspection effort
2. Consistent inspection conditions
3. Automated defect classification
4. Digital inspection records
5. Improved traceability
6. Production-flow integration
7. Modular and scalable architecture

FUTURE SCOPE

1. Larger labelled inspection dataset
2. Improved ML/deep-learning classification
3. Multi-camera inspection
4. Automated physical rejection
5. Industrial production-line integration
6. Remote monitoring and analytics

VIDEO DEMONSTRATION

Watch the working demonstration:

<https://go.screenpal.com/watch/cOjQ6onvu11>

COVER LETTER

Date: 09 August 2026

To,
The A-1 Launchpad 2026 Evaluation Committee
A-1 Fence Products

Subject: Submission of Case Study Solution – Automated Quality Inspection Robot for Wire Mesh Production

Dear Sir/Madam,

We are pleased to submit our solution for the A-1 Launchpad 2026 case study titled “**Automated Quality Inspection Robot for Wire Mesh Production.**”

Our proposed solution focuses on automating the quality inspection of welded wire mesh products through the integration of **mechanical design, sensors and embedded systems, computer vision, machine learning, and automated material handling.**

The proposed system is designed to capture images of manufactured wire mesh, identify and classify defects related to mesh spacing, weld quality, and coating uniformity, generate digital inspection reports, and support integration with the existing production workflow.

The solution has been developed as a multidisciplinary project involving **Mechanical Engineering, Electronics & Communication Engineering**, with each discipline contributing to the design, automation, inspection, and data-processing aspects of the system.

We have included the conceptual design, CAD model, system architecture, prototype development, testing and validation approach, and proposed implementation strategy in this submission. A demonstration video link has also been provided as part of the submission.

We confirm that the submitted work has been developed by our team specifically for this case study and represents our original work.

Team Details

Team Name: A1 Titans

College: Pune Vidyarthi Griha’s College of Engineering, Nashik

Team Members:

1. **Chaitanya Joshi** – Mechanical Engineering
2. **Lavanya Joshi**– Electronics & Communication Engineering

We look forward to presenting our solution and demonstrating its potential for improving the quality, consistency, and efficiency of wire mesh inspection.

Sincerely,

Team A-1 Launchpad 2026

Team Name: A1 Titans

College: Pune Vidyarthi Griha’s College of Engineering, Nashik